4. Both fleshy and dry fruits help in the dispersal of seeds to distant places.
5. In certain cases, fruit may provide nutrition to the developing seedling.
6. Fruits provide source of medicine to human.



- *Lupinus arcticus* (legume family) of Artic Tundra is the oldest viable seed remained dormant for 10,000 years.
- *Pheonix dactylifera* (date palm) of king Herod's palace near dead sea has viable seed for 20,000 years.
- Powdered seeds of *Moringa oleifera* is used to purify water.

4.8 Seed

Do all fruits contain seeds? No, triploid fruits do not. The seed is a fertilized mature ovule which possess an embryonic plant, usually stores food material and has a protective coat. After fertilization, changes occur in various parts of the ovule and transforms into a seed.

4.8.1 Types of Seed

I. Based on the number of cotyledons two types of seeds are recognized.

i. Dicotyledonous seed: Seed with two cotyledons.

ii. Monocotyledonous seed: Seed with one cotyledon.

II. Based on the presence or absence of the endosperm the seed is of two types.

i. Albuminous or Endospermous seed: The cotyledons are thin, membranous and mature seeds have endosperm persistent and nourishes the seedling during its early development. Example: Castor, Sunflower, Maize.

ii. Ex-albuminous or non-endospermous seed: Food is utilized by the developing embryo and so the mature seeds are without endosperm. In such seeds, colyledons store food and become thick and fleshy. Example: Pea, Groundnut.

4.8.2 Significance of Seeds:

- The seed encloses and protects the embryo for next generation.
- It contains food for the development of embryo.
- It is a means for the dispersal of new individuals of the species.
- A seed is a means for perpetuation of the species. It may lie dormant during unfavorable conditions but germinates on getting suitable conditions.
- Seeds of various plants are used as food, both for animals and men.
- They are the basis of agriculture.
- Seeds are the products of sexual reproduction so they provide genetic variation and recombination in a plant.

Activity

Prepare a diet chart to provide balanced diet to an adolescent (a school going child) which includes food items (fruits, vegetable and seeds) which are non - expensive and are commonly available.

Summary

Inflorescence is a group of flowers present on a common stalk. Inflorescence may be classified into 3 types based on position. Inflorescence classified into racemose, cymose, mixed and special type based on the flower arrangement and branching of axis. Flower is a modified shoot and meant for sexual reproduction. Flower has various parts to enhance reproduction. Flower can be explained by its sex and symmetry. Calyx is outermost whorl of flower and many types. Corolla is second whorl of flower and used for pollination. Corolla may be united or free and has various forms in different flowers. Aestivation is arrangement of sepals, petals in bud condition and is of many types. Androecium is the male part of flower and made up of stamens. Stamens contain filament,anther and connective.

Gynoecium is the female part of flower. Ovary, style and stigma are parts of pistil.According to number of carpels it is divided into monocarpellary, bicarpellary etc. It may be apocarpous or syncarpous. Locule number may be one to many.

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